

The *d*- and *f*-Block Elements: Formula Sheet

Class 12 Chemistry – Unit 4 Complete Revision Formulas & Trends

1 Electronic Configurations of *d*- and *f*-Blocks

- General Outer Electronic Configuration of *d*-Block Elements:

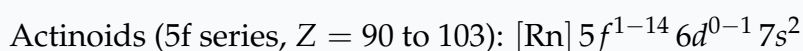
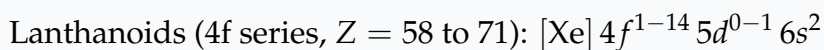
$$(n-1)d^{1-10} ns^{1-2} \quad (\text{Exception: Pd is } 4d^{10}5s^0)$$

Exceptional Stability Configurations in 3d Series:



- General Outer Configuration of *f*-Block Elements (Inner-Transition):

$$\text{General Formula: } (n-2)f^{1-14} (n-1)d^{0-1} ns^2$$



2 Magnetic Moment Formula

- 'Spin-Only' Magnetic Moment (μ):

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Where:

- μ = Magnetic moment in Bohr Magnetons (BM)
- n = Number of unpaired *d*-electrons
- $1 \text{ BM} = \frac{eh}{4\pi m_e} = 9.27 \times 10^{-24} \text{ J} \cdot \text{T}^{-1}$

Values for $n = 1 \rightarrow 5$: $\mu = 1.73, 2.84, 3.87, 4.90, 5.92 \text{ BM}$.

3 Important Oxidizing Compounds: Preparation & Reactions

- 1. Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$):

From Chromite Ore: $4\text{FeCr}_2\text{O}_4 + 8\text{Na}_2\text{CO}_3 + 7\text{O}_2 \rightarrow 8\text{Na}_2\text{CrO}_4 + 2\text{Fe}_2\text{O}_3 + 8\text{CO}_2$

Acidification: $2\text{Na}_2\text{CrO}_4 + 2\text{H}^+ \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + 2\text{Na}^+ + \text{H}_2\text{O}$

pH Interconversion: $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$

Oxidizing Action in Acid: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} \quad (E^\circ = 1.33 \text{ V})$

- 2. Potassium Permanganate (KMnO_4):

From Pyrolusite: $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \rightarrow 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$

Disproportionation: $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$

Acidic Medium: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad (E^\circ = 1.52 \text{ V})$

Neutral / Alkaline: $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^- \quad (E^\circ = 1.69 \text{ V})$

4 Lanthanoid Contraction & Standard Potentials

- Lanthanoid Contraction:** *Steady decrease in atomic and ionic radii of Ln^{3+} (106 pm for $\text{La}^{3+} \rightarrow 86 \text{ pm}$ for Lu^{3+}) due to poor shielding by 4f electrons. Causes 4d and 5d series pairs (e.g., Zr $\approx 160 \text{ pm}$ and Hf $\approx 159 \text{ pm}$) to have almost identical radii.*
- Disproportionation of Copper(I) in Aqueous Solution:**

